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Questions

Q1.

The halogenoalkane 1-chlorobutane reacts under suitable conditions with potassium hydroxide to form the alcohol butan-1-ol.

(a) (i) Name a suitable solvent for the potassium hydroxide in this reaction.

(1)

.....

(ii) State the type and mechanism of this reaction.

(1)

.....

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(iii) Draw the mechanism for this reaction.
Use curly arrows, and show relevant dipoles and lone pairs.

(3)

(b) A student carried out the reaction.

After separation and purification, the mass of butan-1-ol formed was 12.1 g.
The yield of the reaction was 64.0 %.
Calculate the volume of 1-chlorobutane used in the reaction.
Give your answer to an appropriate number of significant figures.
[Density of 1-chlorobutane = 0.886 g cm^{-3}]

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(4)

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(Total for question = 9 marks)

Q2.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Hydrogen peroxide decomposes in the presence of a catalyst.

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

(a) What type of reaction occurs?

(1)

- ☐ **A** displacement
- ☐ **B** disproportionation
- ☐ **C** elimination
- ☐ **D** hydrolysis

(b) In an experiment, the volume of oxygen produced by the decomposition of hydrogen peroxide was measured at various times as the reaction progressed and a graph was plotted.

The initial gradient of the graph was 0.50 $cm^3\ s^{-1}$.

What is the initial rate of decomposition of hydrogen peroxide in mol s^{-1}?

[Molar volume of a gas at r.t.p. = 24 $dm^3\ mol^{-1}$]

(1)

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- ☐ **A** 2.1×10^{-2} mol s^{-1}
- ☐ **B** 4.2×10^{-5} mol s^{-1}
- ☐ **C** 2.1×10^{-5} mol s^{-1}
- ☐ **D** 1.0×10^{-5} mol s^{-1}

(Total for question = 2 marks)

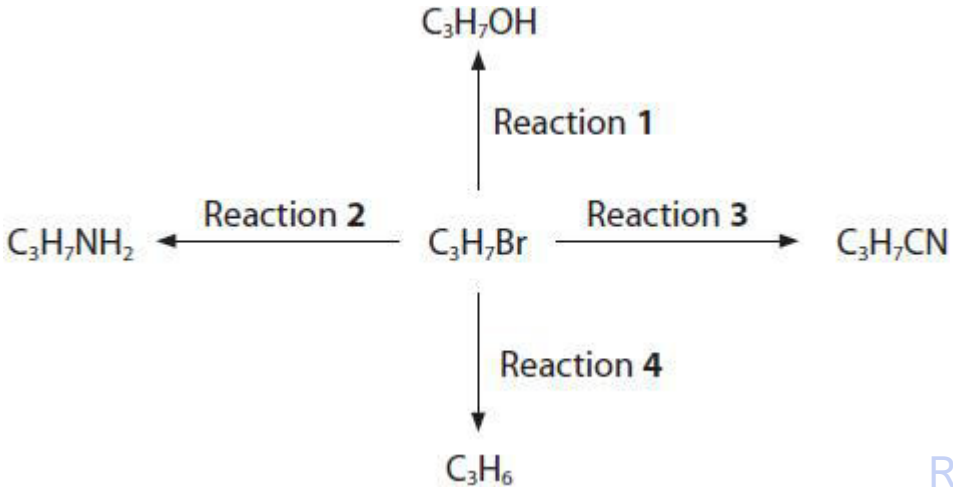
Q3.

Halogenoalkanes are useful reagents in organic synthesis.

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Some reactions of 1-bromopropane are shown.



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(a) Complete the table about these reactions.

(4)

Reaction	Reagent	Solvent	Type of reaction
1	potassium hydroxide		
2	ammonia		
3		ethanol	
4			elimination

(b) Give the IUPAC name of the product of Reaction **3**.

(1)

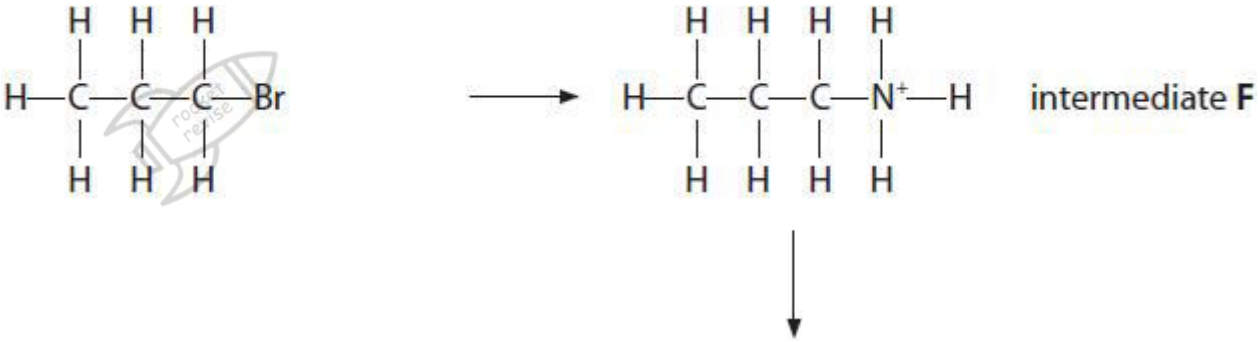
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(c) Complete the mechanism for Reaction **2**.

Show the formation of the intermediate **F** and of the product.
Include curly arrows, and any relevant lone pairs and dipoles.

(3)

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*(d) The relative rates of hydrolysis of a series of halogenoalkanes were determined using the following method:

- five test tubes, each containing 2 cm^3 of ethanol and 2 cm^3 of aqueous silver nitrate, were placed in a water bath at 50 °C
- four drops of a different halogenoalkane were added to each test tube
- the time taken for a precipitate to appear in each test tube was recorded.

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The results are shown.

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Halogenoalkane	Time for ppt to appear / s
2-bromobutane	29
2-chlorobutane	75
2-iodobutane	<1
1-bromobutane	41
2-bromo-2-methylbutane	13

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Explain these results by considering:

- the chemical reaction occurring
- the structures of the halogenoalkanes
- the strengths of the carbon-halogen bonds.

(6)

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(Total for question = 14 marks)

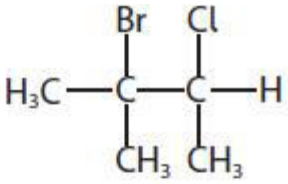
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Q4.

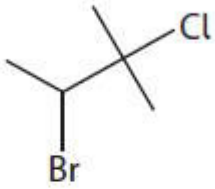
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the structure of 2-bromo-3-chloro-2-methylbutane?

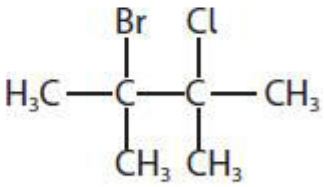
☐ **A**



☐ **B**

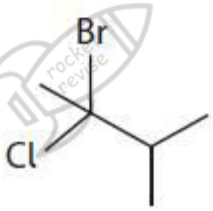


☐ **C**



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☐ **D**



(Total for question = 1 mark)

Q5.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Organic reactions can be classified in different ways.

(a) How should the reaction shown be classified?

(1)



- ☐ **A** addition
- ☐ **B** oxidation
- ☐ **C** polymerisation
- ☐ **D** substitution

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(b) How should the reaction shown be classified?

(1)



- ☐ **A** addition
- ☐ **B** oxidation
- ☐ **C** reduction
- ☐ **D** substitution

(Total for question = 2 marks)

Q6.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Separate samples of some halogenoalkanes were dissolved in ethanol and a few drops of silver nitrate solution added. The faster the reaction of the halogenoalkane, the quicker a precipitate forms.

(a) Which of these halogenoalkanes reacts the **fastest**?

(1)

- ☐ **A** CH_3CHICH_3
- ☐ **B** $CH_3CHBrCH_3$
- ☐ **C** $CH_3CHClCH_3$
- ☐ **D** CH_3CHFCH_3

(b) Which of these halogenoalkanes reacts the **fastest**?

(1)

- ☐ **A** $CH_3CHBrCH(CH_3)CH_3$

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☒ **B** $CH_3CH_2CBr(CH_3)CH_3$
☒ **C** $CH_3CH(CH_2Br)CH_2CH_3$
☒ **D** $CH_3CH_2CH_2CH_2CH_2Br$

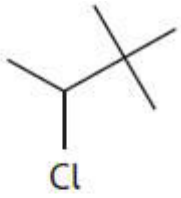
(Total for question = 2 marks)

Q7.

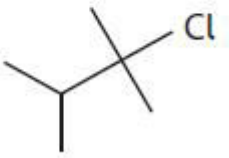
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which structure represents a primary halogenoalkane?

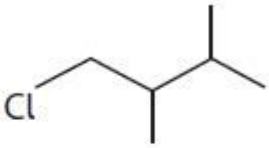
☒ **A**



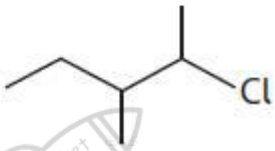
☒ **B**



☒ **C**



☒ **D**



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(Total for question = 1 mark)

Q8.

This question is about halogenoalkanes.

(a) The rates of hydrolysis of 1-chloropropane, 1-bromopropane and 1-iodopropane in reactions with aqueous silver nitrate solution were compared.

(i) State what would be measured in the experiment to compare the rates of hydrolysis.

(1)

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(ii) State which of these halogenoalkanes would hydrolyse the fastest.
Justify your answer.

(2)

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(b) The equation for the hydrolysis of 1-chloropropane by aqueous hydroxide ions is shown.

$$CH_3CH_2CH_2Cl + OH^- \rightarrow CH_3CH_2CH_2OH + Cl^-$$

Give the mechanism for this hydrolysis.
Include curly arrows, and relevant dipoles and lone pairs.

(2)

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(c) The boiling temperatures of some halogenoalkanes are shown.

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Halogenoalkane	Boiling temperature / °C
1-chloropropane	47
1-bromopropane	71
1-iodopropane	103

Explain the trend in boiling temperature of these halogenoalkanes by comparing the intermolecular forces involved.
Detailed explanations of the forces involved are not required.

(4)

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(d) In another experiment, 2-bromobutane is heated with ethanolic potassium hydroxide and an elimination reaction occurs.

Draw the **skeletal** formulae of the three possible organic products, giving their names.

(3)

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Skeletal formula	Name

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(Total for question = 12 marks)

Q9.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

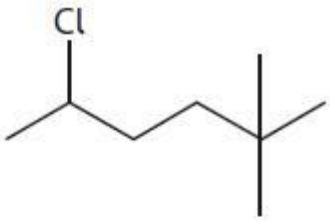
What is the skeletal formula of 2-chloro-4,4-dimethylhexane?

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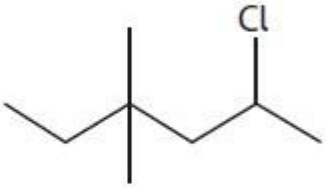
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☐ **A**

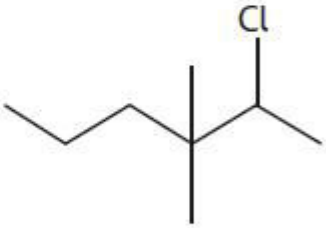


☐ **B**

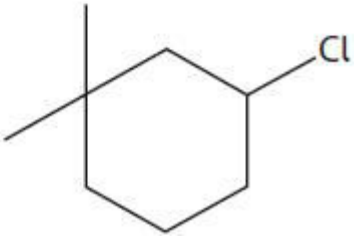


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☐ **C**



☐ **D**



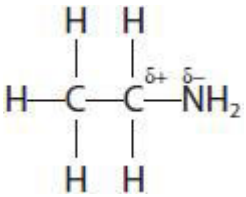
(Total for question = 1 mark)

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Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structure of ethylamine is shown.



What type of reagent will attack the Cδ+ atom?

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- ☒ **A** electrophile
- ☒ **B** free radical
- ☒ **C** nucleophile
- ☒ **D** oxidising agent

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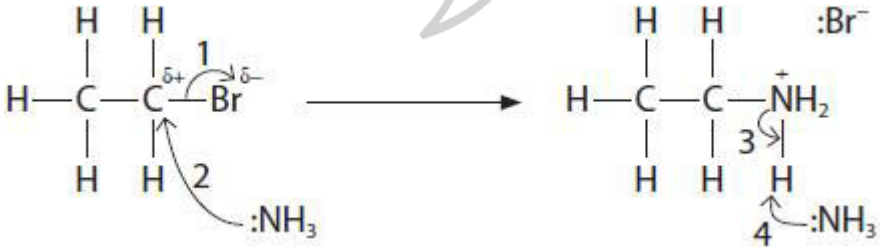
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(Total for question = 1 mark)

Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Part of a reaction mechanism is shown.



Which curly arrow in the mechanism is **not** correct?

- ☒ **A** arrow 1
- ☒ **B** arrow 2
- ☒ **C** arrow 3
- ☒ **D** arrow 4

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(Total for question = 1 mark)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Butanenitrile can be formed by reacting 1-bromopropane with potassium cyanide.

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Which is the correct mechanism and type of reaction that occurs?

- ☒ **A** electrophilic addition
- ☒ **B** nucleophilic addition
- ☒ **C** electrophilic substitution
- ☒ **D** nucleophilic substitution

(Total for question = 1 mark)

Q13.

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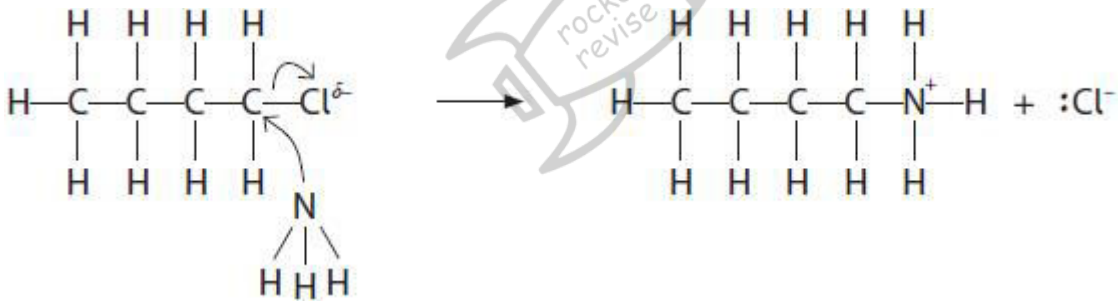
Ammonia reacts with 1-chlorobutane.

(a) State the type and mechanism of this reaction.

(2)

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(b) A student drew the first step of the mechanism for the reaction.



(i) Identify **two** omissions in the student's mechanism.

(2)

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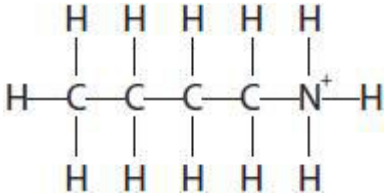
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(ii) To obtain butylamine, sodium hydroxide solution is added.
Complete the next step of the mechanism to form butylamine, showing curly arrows, relevant lone pairs and the reaction products.

(3)



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(c) The reactions of ammonia and of hydroxide ions with halogenoalkanes are similar.

Compare the rate of reaction of ammonia with 1-chlorobutane and with 2-bromo-2-methylpropane. Justify your answer.

(3)

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(Total for question = 10 marks)

Q14.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When 1-chlorobutane is mixed with ethanol and aqueous silver nitrate, a reaction occurs.

Which species is the nucleophile in this reaction?

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- ☐ **A** ethanol
- ☐ **B** nitrate ions
- ☐ **C** silver ions
- ☐ **D** water

(Total for question = 1 mark)

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Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

$CH_3CH_2CHBrCH_2CH_2CH_3$ is heated with ethanolic potassium hydroxide.

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How many **alkene** products are possible?

- ☐ **A** one
- ☐ **B** two
- ☐ **C** three
- ☐ **D** four

(Total for question = 1 mark)

Q16.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which compound reacts with ammonia to form $CH_3CH(NH_2)CH_3$?

- ☐ **A** 1-chloropropane
- ☐ **B** 2-chloropropane
- ☐ **C** propane
- ☐ **D** propene

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(Total for question = 1 mark)

Q17.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Separate solutions of 1-chloropropane and 1-bromopropane in ethanol are warmed with aqueous silver nitrate.

Why does the formation of a precipitate take longer with 1-chloropropane?

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☒ **A** 1-chloropropane is less soluble than 1-bromopropane
☒ **B** the C—Cl bond is stronger than the C—Br bond
☒ **C** the C—Cl bond is more polar than the C—Br bond
☒ **D** silver chloride is more soluble than silver bromide

(Total for question = 1 mark)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Silver nitrate in aqueous ethanol is added separately to four halogenoalkanes.

Which would form a silver halide precipitate in the **shortest** time?

☒ **A** $(CH_3)_3Cl$
☒ **B** $(CH_3)_3CCl$
☒ **C** $CH_3CH_2CH_2CH_2I$
☒ **D** $CH_3CH_2CH_2CH_2Cl$

(Total for question = 1 mark)

Q19.

This question is about sodium hydroxide.

(a) (i) Write an **ionic** equation for the neutralisation reaction between aqueous sodium hydroxide and hydrochloric acid. State symbols are not required.

(1)

(ii) State what is meant by standard enthalpy change of neutralisation, $\Delta_{neut}H^{\ominus}$.

(2)

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(b) A student carried out an investigation to determine the enthalpy change of neutralisation of aqueous

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sodium hydroxide by hydrochloric acid.

Method

- separate 25.0 cm^3 samples of 0.80 mol dm^{-3} sodium hydroxide and 0.80 mol dm^{-3} hydrochloric acid were left to reach room temperature

- after two minutes, the solutions were mixed in a copper calorimeter and the temperature was noted at 30 s intervals.

(i) Use the graph shown to determine the maximum temperature change, ΔT, in this experiment. You **must** show your working on the graph.

(2)

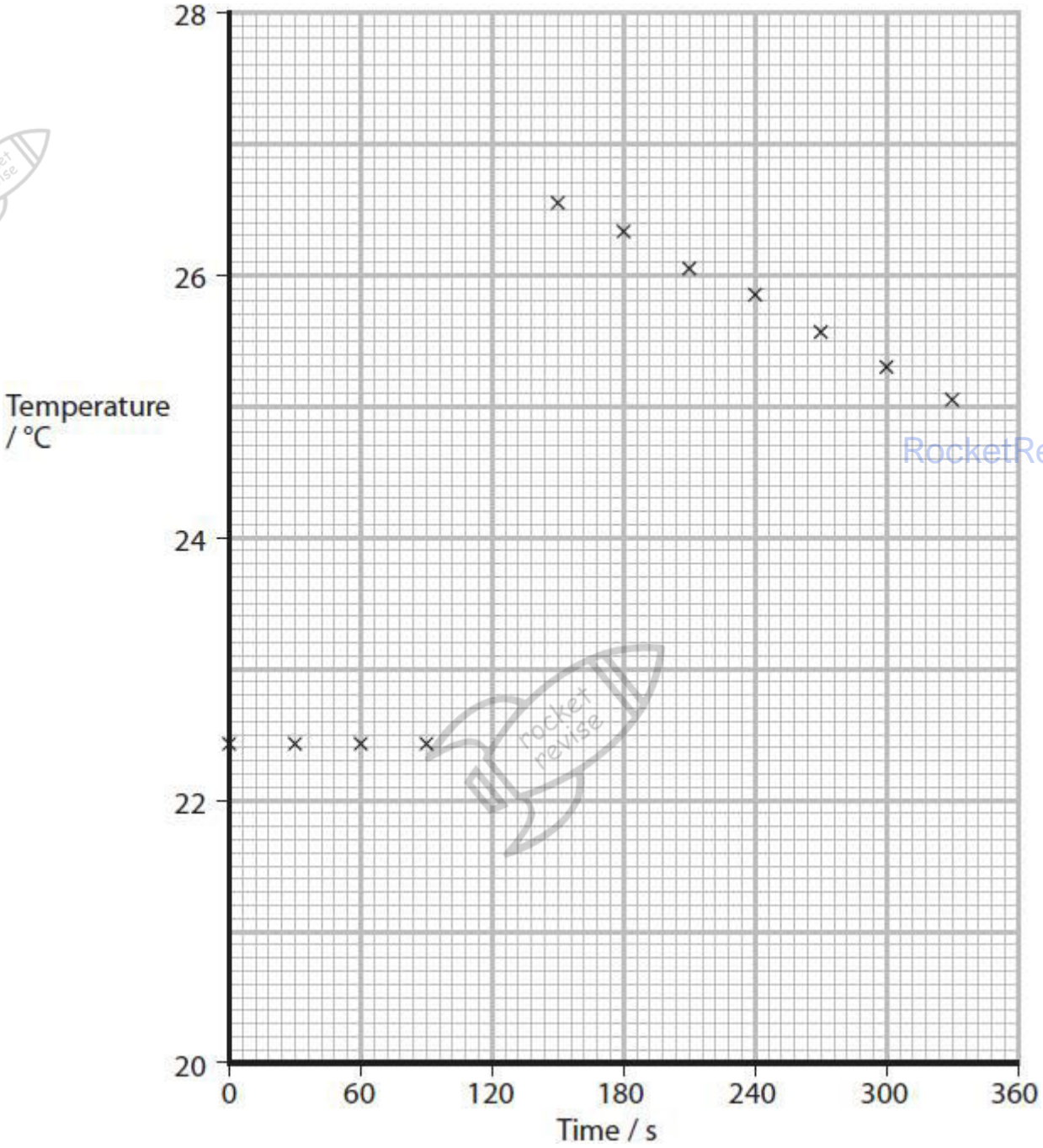
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ΔT .....

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(ii) Calculate the enthalpy change of neutralisation using your answers to (a) and (b)(i). Give a sign and units with your answer.

Assume: no energy is used to heat the container.
the specific heat capacity of the solution = 4.2 J $°C^{-1}$ g^{-1}.
the densities of the solutions of NaOH and HCl are 1.0 g cm^{-3}.

(3)

(iii) Explain how, if at all, the enthalpy change of neutralisation obtained in (b)(ii) would differ if the heat capacity of the calorimeter was included in the calculation.

(2)

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(c) Aqueous sodium hydroxide reacts with 1-bromopropane to produce propan-1-ol.

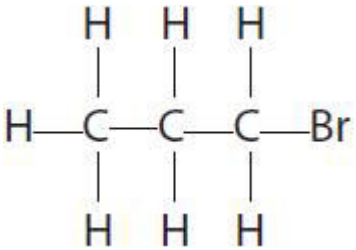
(i) State the type and mechanism of this reaction.

(1)

.....

(ii) Complete the mechanism for this reaction.
Include curly arrows, and relevant lone pairs and dipoles.

(3)



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(iii) Under different conditions, sodium hydroxide reacts with 1-bromopropane to form propene.
Name the type of reaction and a suitable solvent.

(2)

Type of reaction

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Suitable solvent

.....

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(Total for question = 16 marks)

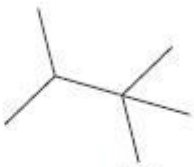
Q20.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The structures of three bromoalkanes **P**, **Q** and **R** are shown.

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P	**Q**	**R**

Which of these bromoalkanes may be classified as tertiary?

- ☒ **A** **P** only
- ☒ **B** **Q** only
- ☒ **C** **R** only
- ☒ **D** **Q** and **R** only

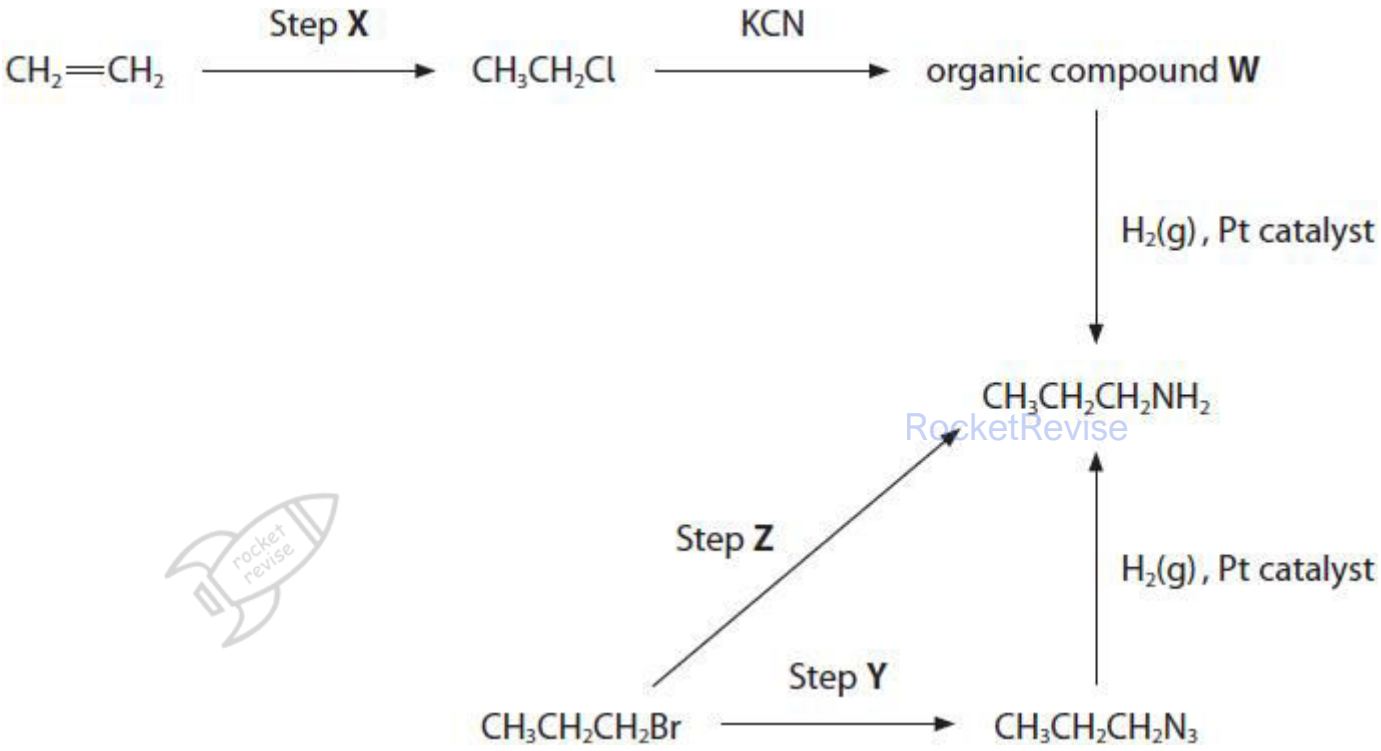
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(Total for question = 1 mark)

Q21.

This question is about the synthesis of propylamine, $CH_3CH_2CH_2NH_2$.

Three routes for the synthesis of $CH_3CH_2CH_2NH_2$ are shown.



(a) Identify, by name or formula, the reagent used in Step **X**.

(1)

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(b) Give the structure of organic compound **W**.

(1)

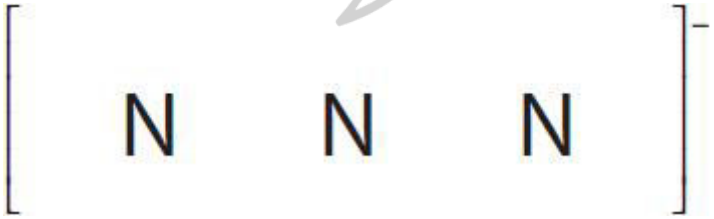
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(c) The reagent used in Step **Y** is potassium azide, KN_3.

This is a source of the azide ion, N_3^-, which acts as a nucleophile.

(i) Complete a possible dot-and-cross diagram for N_3^-.
Show outer electrons only.

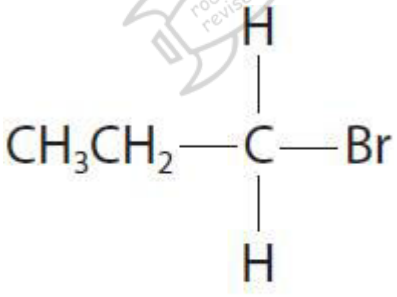
(2)



(ii) Complete the mechanism for Step **Y**.
Include curly arrows, and any relevant lone pairs and dipoles.

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(3)



N_3^-

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(d) Step **Z** is carried out by reacting ammonia with 1-bromopropane.

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(i) Give the conditions for this reaction.

(2)

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(ii) Suggest why the yield of $CH_3CH_2CH_2NH_2$ obtained using Step **Z** is low.

(1)

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(Total for question = 10 marks)

Q22.

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Toilet Cleaners

Products used to clean toilets work in two ways: removing the limescale that collects in the toilet bowl and disinfecting the bowl.

Limescale contains calcium carbonate which is removed by reaction with acids. The strongest acid used in toilet cleaners is hydrochloric acid, although weaker acids such as methanoic acid are also used.

Disinfectants are essentially chlorine dissolved in aqueous sodium hydroxide. The effective disinfectant in these solutions is the chlorate(I) ion, ClO^-.

(a) State **one** safety precaution that you should take when using a toilet cleaner containing hydrochloric acid.

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(1)

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(b) The equation for the reaction of calcium carbonate with hydrochloric acid is shown.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

(i) Write the **ionic** equation for this reaction.
State symbols are not required.

(1)

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(ii) A 750 cm^3 bottle of a toilet cleaner contains 85.0 g of hydrochloric acid.
Calculate the maximum mass of calcium carbonate limescale that could be removed by 50.0 cm^3 of this toilet cleaner.

(4)

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(c) Chlorine gas reacts with water.

$$Cl_2(g) + H_2O(l) \rightleftharpoons HClO(aq) + HCl(aq)$$

(i) State the classification of this reaction. Justify your answer in terms of the relevant oxidation numbers.

(3)

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(ii) When 5.00 cm^3 of a disinfectant reacts with excess hydrogen peroxide, 113 cm^3 of oxygen is produced, measured at room temperature and pressure.

$$H_2O_2(aq) + NaClO(aq) \rightarrow H_2O(l) + NaCl(aq) + O_2(g)$$

Calculate the concentration, in g dm^{-3}, of sodium chlorate(I) in the disinfectant.
Give your answer to **two** significant figures.

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(4)

[Molar volume of gas at r.t.p. = 24 000 cm^3 mol^{-1}]

(iii) Commercial disinfectants are often made more viscous by adding propane-1,2,3-triol to the aqueous solution.
Suggest how propane-1,2,3-triol makes disinfectants more viscous.

(2)

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(d) Toilet cleaners that remove limescale should not be mixed with chlorine-based disinfectants.

Explain, using the relevant equation, why these two types of toilet cleaner should not be mixed.

(2)

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(e) Borax is another cleaning agent used to remove limescale.

Borax is a compound of sodium, boron and oxygen.
Borax crystals contain water of crystallisation.
The percentage composition by mass of some borax crystals is
H = 5.2 % B = 11.3 % O = 71.4 % Na = 12.1 %
Determine the empirical formula of the borax crystals and hence their formula.

(4)

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(Total for question = 21 marks)

Q23.

1-bromobutane reacts with alkali:

$CH_3\text{-}CH_2\text{-}CH_2\text{-}CH_2Br + OH^- \rightarrow CH_3\text{-}CH_2\text{-}CH_2\text{-}CH_2OH + Br^-$

The mechanism and type of reaction is:

(1)

- ☒ **A** electrophilic addition
- ☒ **B** electrophilic substitution
- ☒ **C** nucleophilic addition
- ☒ **D** nucleophilic substitution

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(Total for question = 1 mark)

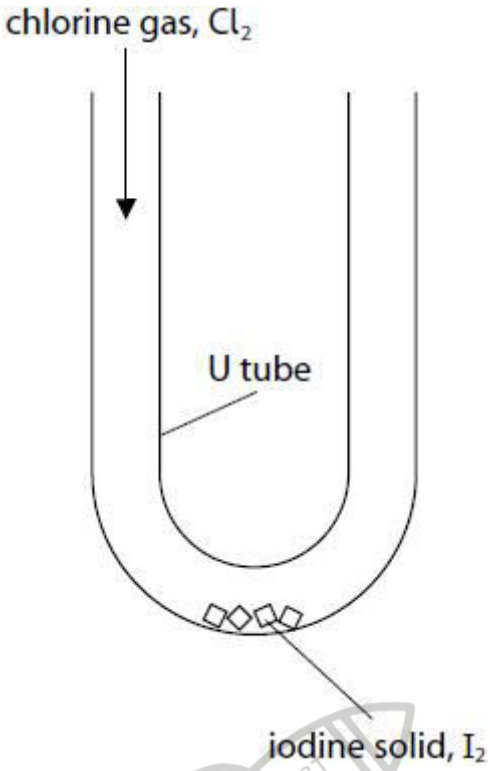
Q24.

Iodine monochloride, ICl, is a covalent compound produced by the reaction of iodine with chlorine. Iodine monochloride is a dark brown liquid at room temperature.

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The equipment shown can be used to pass chlorine over solid iodine to produce iodine monochloride.



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When excess chlorine is passed through the U tube, the iodine monochloride reacts to produce iodine trichloride in an equilibrium reaction.

(a) Write a chemical equation for the reaction of iodine with chlorine to produce iodine monochloride. Include state symbols.

(2)

(b) The iodine monochloride molecule has a permanent dipole. Complete the following table using the electronegativity data from your Data Booklet and hence show the dipole on the diagram of the iodine monochloride molecule.

(1)

Element	Electronegativity
Cl	
I	

I — Cl

(c) Iodine monochloride reacts with propene to form two isomeric products. This is an addition reaction

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that is similar to the reaction of propene with hydrogen halides.

(i) Draw the skeletal formulae of both isomers.

(2)

(ii) Explain which of these isomers is the major product.

(3)

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(d) The equation for the reaction between iodine monochloride and chlorine is:

$$\underset{\text{brown liquid}}{ICl(l)} + Cl_2(g) \rightleftharpoons \underset{\text{yellow solid}}{ICl_3(s)}$$

(i) State and justify **one** precaution that must be taken when preparing iodine trichloride.

(2)

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(ii) Give the oxidation number of iodine in both iodine-containing compounds in the equilibrium.

(1)

I in ICl

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I in ICl_3

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(iii) Iodine trichloride can also be made by reacting potassium chlorate(V) with iodine in hydrochloric acid. The equation for the reaction is

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$KClO_3 + I_2 + 6HCl \rightarrow KCl + 2ICl_3 + 3H_2O$

By considering oxidation numbers for chlorine, explain whether or not this reaction is a disproportionation.

(2)

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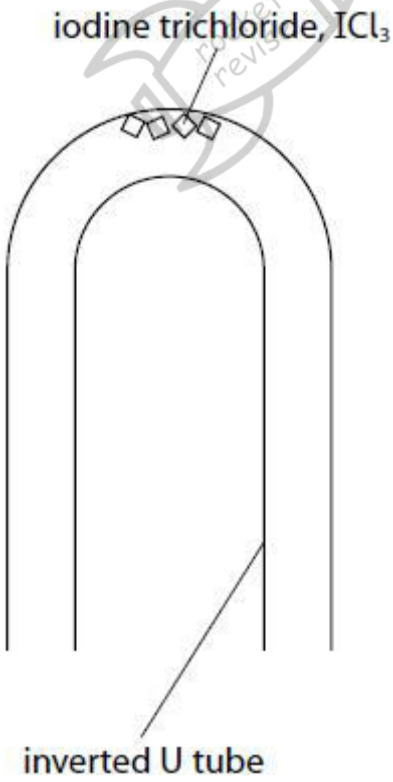
(e) Chlorine gas has a molar volume of 24 000 cm^3 mol^{-1} under the conditions used in this reaction.

(i) Show that the density of chlorine gas is approximately 3 g dm^{-3}.

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(2)

(ii) Air has an average density of 1.25 g dm^{-3}. If the U-tube used in part (d) is inverted, as shown in the diagram, the solid yellow iodine trichloride produced in the equilibrium reaction turns to a brown liquid.



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Explain this observation.

(3)

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(f) A mass of 0.64 g of iodine reacted with fluorine to form 1.31 g of a fluoride of iodine.

Calculate the empirical formula of this compound of iodine and fluorine.

(2)

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(Total for question = 20 marks)

Q25.

Which compound is hydrolysed most rapidly?

(1)

- ☒ **A** 1-chloropropane
- ☒ **B** 1-chlorobutane
- ☒ **C** 2-chloro-2-methylpropane
- ☒ **D** 2-chlorobutane

(Total for question = 1 mark)

Q26.

Give the systematic name for the following molecule.

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(1)

- ☒ **A** *E*-1-bromo-2-methylbut-2-ene
- ☒ **B** *E*-2-methyl-1-bromobut-1-ene
- ☒ **C** *E*-1-bromo-3-methylpent-2-ene
- ☒ **D** *E*-1-bromo-2-methylbut-1-ene

(Total for question = 1 mark)

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Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Silver nitrate, in aqueous ethanol, is added separately to four halogenoalkanes.

Which would form a silver halide precipitate in the shortest time?

- ☒ **A** 1-bromobutane
- ☒ **B** 1-chlorobutane
- ☒ **C** 1-fluorobutane
- ☒ **D** 1-iodobutane

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(Total for question = 1 mark)

Q28.

The elements of Group 7, the halogens, have many important uses.

Fluorine is used to make sulfur hexafluoride, SF_6, a gas which is used as an insulator in electrical equipment.

Chlorine is used as a bleach and as a disinfectant in water treatment.

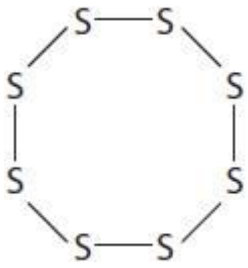
Bromine is used to make organic chemicals that are used as fire retardants.

Iodine has been used in the catalyst in the manufacture of ethanoic acid.

(a) SF_6 can be made from the reaction of sulfur, S_8, with fluorine.

$$S_8(s) + 24F_2(g) \rightarrow 8SF_6(g) \qquad \Delta_r H = -9672\ kJ\ mol^{-1}$$

The structure of $S_8(s)$ is shown.



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Bond enthalpy data

Bond	Mean bond enthalpy / kJ mol^{-1}
S—S	268
F—F	151

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(i) Calculate the mean bond enthalpy, in kJ mol^{-1}, for a single S—F bond.

(3)

(ii) Give **two** reasons why your answer to (a)(i) is unlikely to accurately represent the true S—F bond enthalpy in SF_6.

(2)

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(iii) SF_6 is a potent greenhouse gas.
1.00 g of SF_6 absorbs the same amount of infrared radiation as 23.9 kg of CO_2.
Calculate the number of molecules of CO_2 , to the nearest whole number, that absorb the same amount of infrared radiation as one molecule of SF_6 in the atmosphere.

(3)

(b) Bleach, NaClO(aq), is formed by the reaction of chlorine with cold, dilute aqueous sodium hydroxide.

$$Cl_2(g) + 2NaOH(aq) \rightarrow NaClO(aq) + NaCl(aq) + H_2O(l)$$

The following species and equilibria are present in solutions of NaClO(aq).

Equilibrium 1 $HClO(aq) \rightleftharpoons H^+(aq) + ClO^-(aq)$

Equilibrium 2 $HClO(aq) + Cl^-(aq) + H^+(aq) \rightleftharpoons Cl_2(aq) + H_2O(l)$

Equilibrium 3 $Cl_2(aq) \rightleftharpoons Cl_2(g)$

Household bleach is stored at a pH between 11 and 13.
Household bleach should not be stored near strong acid.
(i) Give the formula of the chlorine-containing species present at highest concentration in household bleach at pH 12.

(1)

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(ii) Explain why mixing strong acid with bleach is dangerous.
Justify your answer using the equilibria shown.

(3)

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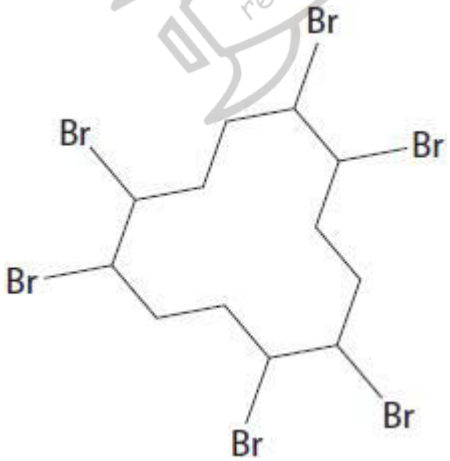
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(iii) On heating, NaClO disproportionates to form sodium chloride and sodium chlorate(V).
Give the equation for this disproportionation reaction.
State symbols are **not** required.

(2)

(c) The structure of a fire retardant is shown.



Calculate the percentage by mass of bromine in this fire retardant.
Give your answer to an appropriate number of significant figures.

(2)

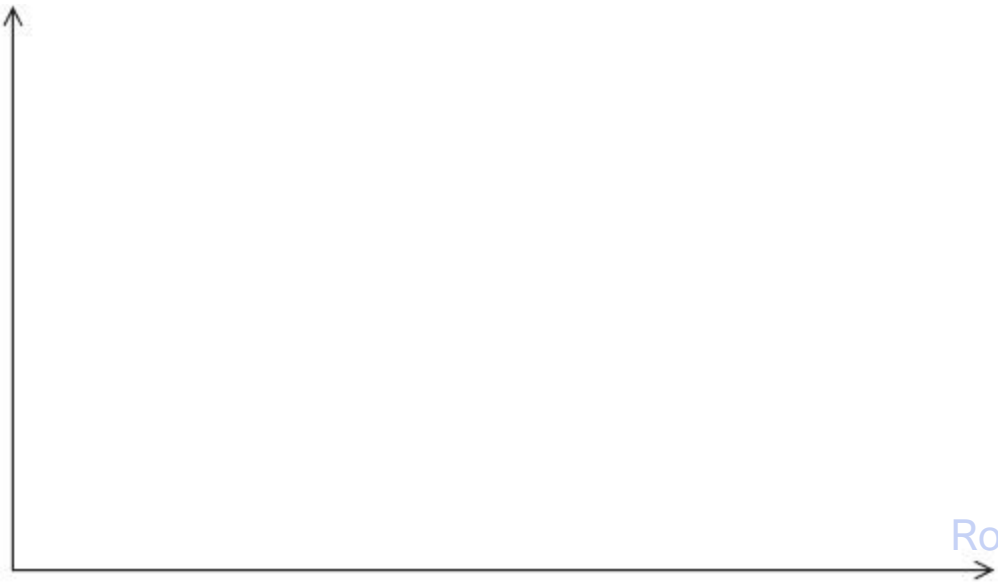
(d) The industrial manufacture of ethanoic acid first used an iodine-based catalyst.

Explain, with the aid of a labelled Maxwell–Boltzmann distribution, how a catalyst increases the rate of a chemical reaction.

(4)

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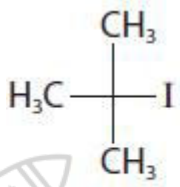
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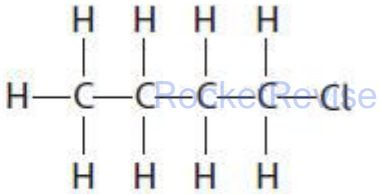
(Total for question = 20 marks)

Q29.

This question is about two halogenoalkanes, **A** and **B**, which have the structures shown.



A



B

(a) (i) Name compound **A**.

(1)

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(ii) Draw the **skeletal** formula of **B**.

(1)

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(b) (i) Describe what you would observe when separate samples of **A** and **B** are warmed with silver nitrate dissolved in aqueous ethanol.

(2)

Observation with **A**

.....

Observation with **B**

.....

(ii) Give **two** reasons why **A** reacts more quickly than **B** when warmed with silver nitrate dissolved in aqueous ethanol.

(2)

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(c) (i) Write an equation for the reaction of **B** with an alcoholic solution of potassium cyanide and name the organic product.

(2)

Equation:

Name

.....

(ii) Suggest why this type of reaction is important in organic synthesis.

(1)

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(Total for question = 9 marks)

Q30.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Chloroethane reacts with aqueous potassium hydroxide to produce ethanol.

(a) What is the role of the hydroxide ion in this reaction?

(1)

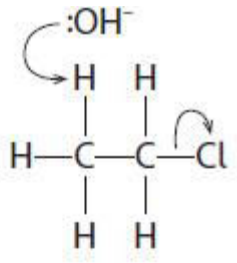
- ☐ **A** reducing agent
- ☐ **B** nucleophile
- ☐ **C** electrophile
- ☐ **D** base

(b) Which mechanism shows the correct movement of electrons in this reaction?

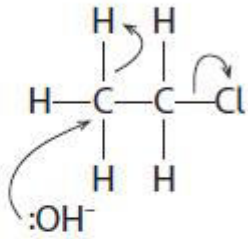
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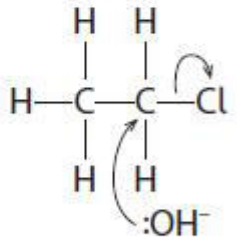
☐ **A**



☐ **B**

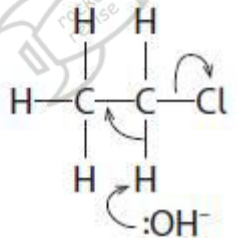


☐ **C**



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☐ **D**



(Total for question = 2 marks)

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Q31.

This question is about the reactions of halogenoalkanes.

(a) When halogenoalkanes are heated with potassium hydroxide, the products depend on the conditions.

(i) State what condition, other than a suitable temperature and concentration of potassium hydroxide, would give a high yield of but-1-ene and but-2-ene from 2-chlorobutane.

(1)

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(ii) Give the type of reaction occurring in (a)(i).

(1)

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(b) (i) State what condition, other than a suitable temperature and concentration of potassium hydroxide, would give a high yield of butan-1-ol from 1-chlorobutane.

(1)

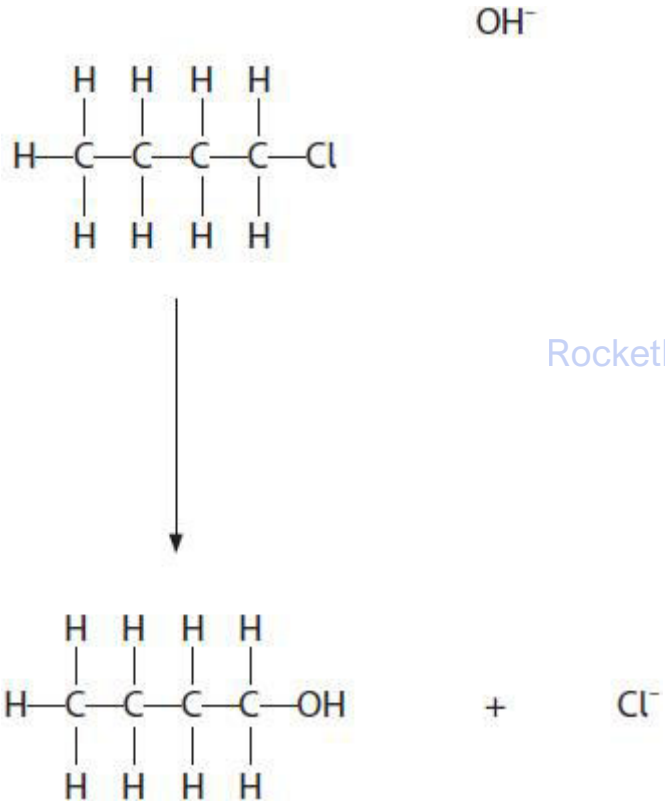
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(ii) Complete the mechanism for the reaction of potassium hydroxide with 1-chlorobutane to form butan-1-ol.

Use curly arrows and show relevant dipoles and lone pairs.

(2)



(iii) Give **two** reasons why, under the conditions used in (b)(ii), the reaction between potassium hydroxide and 2-iodo-2-methylpropane is faster than the reaction between potassium hydroxide and

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1-chlorobutane.

(2)

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(Total for question = 7 marks)

Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

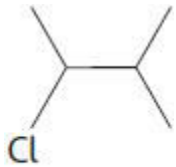
An alcohol can be attacked by a nucleophile because

- ☒ **A** the C—O bond is weak
- ☒ **B** the C—O bond is polar
- ☒ **C** the O—H bond can form hydrogen bonds
- ☒ **D** the O—H bond is polar

(Total for question = 1 mark)

Q33.

Compound **X** is a halogenoalkane.



X

(a) Name compound **X**.

(1)

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(b) The reaction of **X** with hot, concentrated ethanolic KOH produces two isomers.

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(i) Name this type of reaction.

(1)

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(ii) Draw the structure of the two isomers.

(2)

Isomer 1

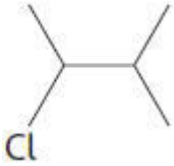
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Isomer 2

(c) (i) Complete the reaction mechanism for compound **X** reacting with cold, dilute aqueous potassium hydroxide.

Include curly arrows and relevant lone pairs and dipoles.

(3)



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(ii) Compound **Y** also reacts with cold, dilute aqueous potassium hydroxide.

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Y

Explain how the rate of reaction will compare with that of compound **X**.

(2)

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(d) State the role of the hydroxide ion in each of the reactions (b) and (c).

(2)

Role in (b) .....

Role in (c) .....

(Total for question = 11 marks)

Q34.

This question is about nitrogen, hydrogen and their compounds.

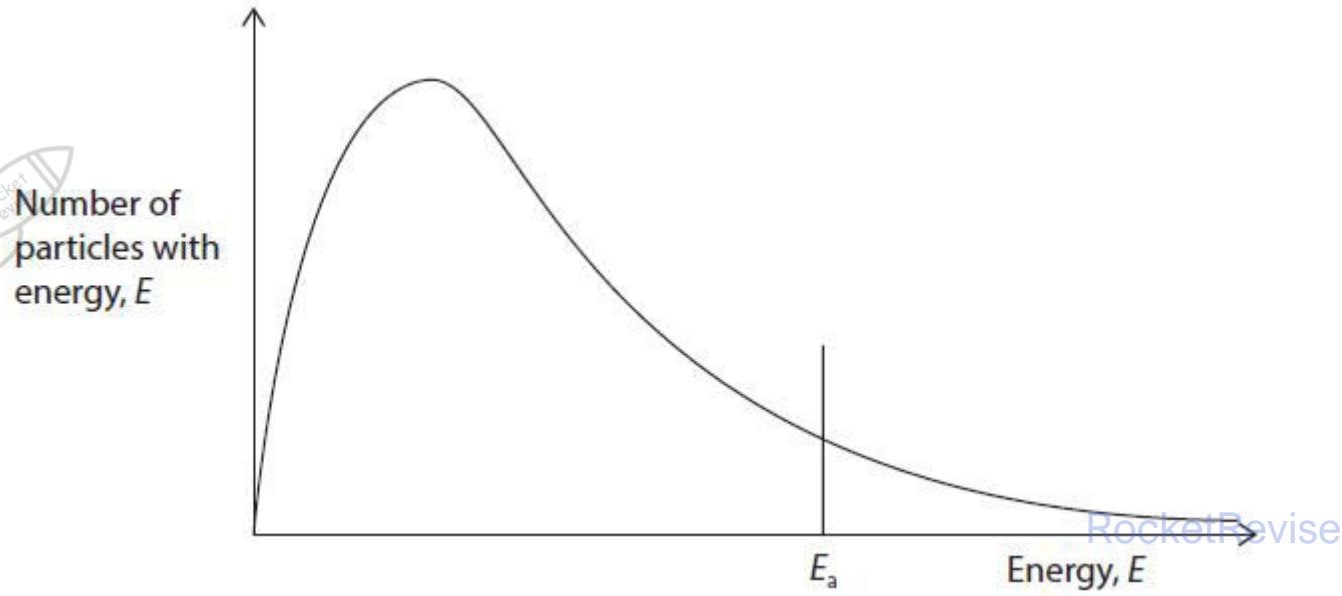
(a) Ammonia is produced in the Haber process.

Nitrogen and hydrogen react at a pressure of 200 atmospheres and a temperature of 725K, using an iron catalyst.
Explain how a catalyst increases the rate of a chemical reaction, using the Maxwell−Boltzmann distribution shown, and the collision theory.

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(3)

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(b) Storing and transporting hydrogen can be difficult due to its low energy density.

This has led to the use of various hydrides which contain a high proportion of hydrogen and are easier to liquefy and transport.

(i) Explain the differences in the boiling temperatures of the Group 5 hydrides shown.
Detailed descriptions of the intermolecular forces are not required.

Hydride	Boiling temperature / K
NH_3	240
PH_3	186
AsH_3	212
SbH_3	256

(4)

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(ii) Hydrogen has been proposed as an alternative fuel to petrol.
Explain why hydrogen is a more environmentally friendly fuel than alkanes.

(3)

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(c) Ethylamine may be prepared from the reaction between ammonia and bromoethane.

(i) State the conditions for this reaction.

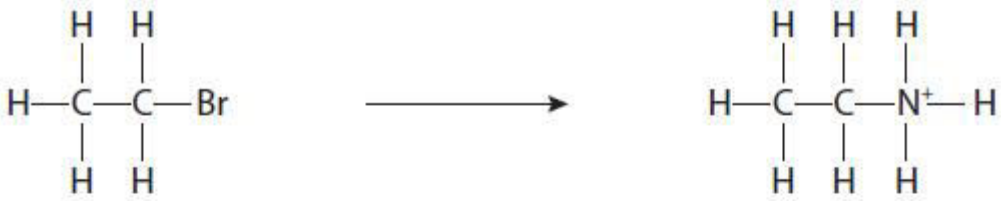
(2)

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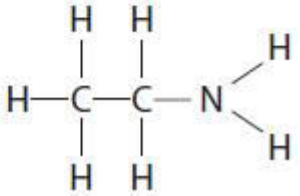
(ii) Complete the mechanism for this reaction.
Include curly arrows, and relevant dipoles and lone pairs.

(3)



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(iii) Further substitution of the ethylamine molecule may occur if the halogenoalkane is in excess. State **two** features of the ethylamine molecule that make this possible.

(2)

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(Total for question = 17 marks)

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Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

Halogenoalkanes, R—X, react when warmed with aqueous silver nitrate in the presence of ethanol. The rates of reaction of 1-chlorobutane, 1-bromobutane and 1-iodobutane were compared.

(a) The rate was fastest for 1-iodobutane and slowest for 1-chlorobutane.

Which is the **best** explanation for this result?

(1)

- ☐ **A** the bond strength of the hydrogen halide molecule decreases going down the group
- ☐ **B** the carbon-halogen bond polarity decreases going down the group
- ☐ **C** the first ionisation energy of the halogen decreases going down the group
- ☐ **D** the carbon-halogen bond strength decreases going down the group

(b) What is observed when 1-bromobutane is warmed with aqueous silver nitrate in the presence of ethanol?

(1)

- ☐ **A** an orange solution
- ☐ **B** a cream precipitate soluble in dilute aqueous ammonia
- ☐ **C** a cream precipitate insoluble in dilute aqueous ammonia
- ☐ **D** a yellow precipitate insoluble in dilute aqueous ammonia

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(c) What is the mechanism of these hydrolysis reactions to form alcohols?

(1)

- ☐ **A** electrophilic addition
- ☐ **B** elimination
- ☐ **C** free radical substitution
- ☐ **D** nucleophilic substitution

(Total for question = 3 marks)

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